Enrolment No.: _____

Darshan UNIVERSITY
योगः कर्मसु कौशलम्

School of Computer Science, Darshan University
MCA | Sem-2 | Summer-2022

Course Code	:	2105CS202	**Date**	:	01.06.2022
Course Name	:	Python for Data Science	**Duration**	:	150 Minutes
			Total Marks	:	70

Instructions:
1. Attempt all the questions.
2. Figures to the right indicates maximum marks.
3. Make suitable assumptions wherever necessary.

Q.1	**(A)**	List and explain steps of Data Science Pipeline.	**4**
	(B)	Explain %timeit command with example.	**3**
		OR	
		Explain %run Command with example.	
	(C)	What is NumPy? Explain arange, zeros, ones, eye, linspace, rand and randint functions with examples.	**7**
		OR	
		Explain conditional selection with DataFrame with example.	
Q.2	**(A)**	Compare Numpy and Pandas.	**4**
	(B)	Explain join function in pandas with example.	**3**
		OR	
		Explain merge function in pandas with example.	
	(C)	Explain NetworkX library with example.	**7**
		OR	
		Explain different types of graphs available in seaborn.	
Q.3	**(A)**	Explain Multi-Index DataFrame in pandas with example.	**4**
	(B)	Explain Axes, Ticks and Grid in Matplotlib with example.	**3**
		OR	
		Explain Labels, Annotation and Legends in Matplotlib.	
	(C)	Write a python program that performs following tasks using NumPy 1) Create NumPy array of 200 random integer numbers from 1 to 200. 2) Display the standard deviation. 3) Display the sum of all the elements. 4) Display all the elements which are less than 10 from array. 5) Display only odd elements from array. 6) Reshape the array into 4 * 50.	**7**

7) Display the sum of all Columns.

OR

Consider following Restaurant_order.csv

order_id	quantity	item_name	choice_description	item_price

Using pandas perform the following tasks.

1) Import the dataset from 'http://abc.com/ Restaurant_order.csv' and assign it to a variable called 'rsorder'
2) Print the name of all the columns.
3) Find out numbers of rows in the dataset.
4) Which was the most-ordered item?
5) For the most-ordered item, how many items were ordered?
6) Create a lambda function and change the type of item price
7) How much was the revenue for the period in the dataset?

Q.4 **(A)** Write a Pandas program to drop the columns where at least one element is missing in a given DataFrame. **4**

	Rollno	Enrollment	Name
0	101	1111	NaN
1	102	1122	NaN
2	103	1133	xyz

(B) Discuss the Classes available in Scikit-learn library. **3**

OR

Compare Supervised and Unsupervised learning.

(C) What is clustering? Explore K-mean on sample data (random X and Y values) to create 3 clusters and print center of each cluster. **7**

OR

What is Exploratory Data Analysis (EDA)? Explain it with titanic datasets

Q.5 **(A)** Explain TF-IDF with example. **4**

(B) Explain Generalization, Overfitting, and Underfitting. **3**

OR

Explain the confusion matrix with an example.

(C) What is Linear Regression? Write a python program to predict future price of iPhone. Consider Columns of CSV file as follow. **7**
iphoneNumber, model_name, price

OR

What is Classification? Write a python program to classify iris flowers. Use K - nearest Neighbors algorithm for classification. For training of model use inbuild iris dataset available in sklearn.

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